

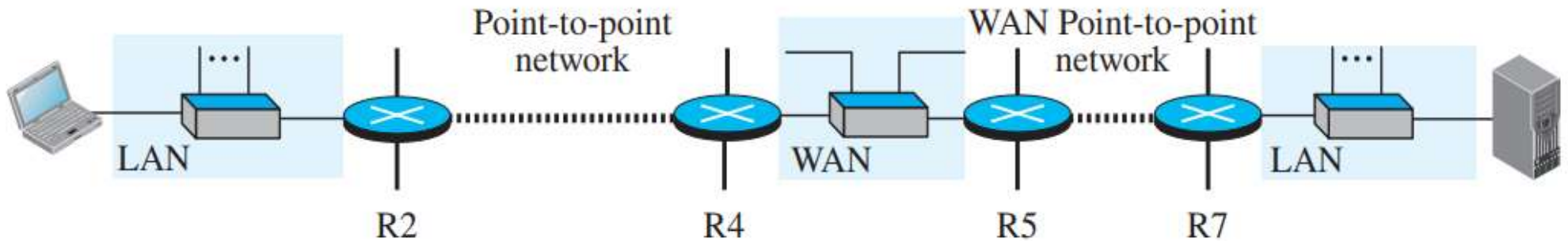
Module No. 3

Data Link Layer

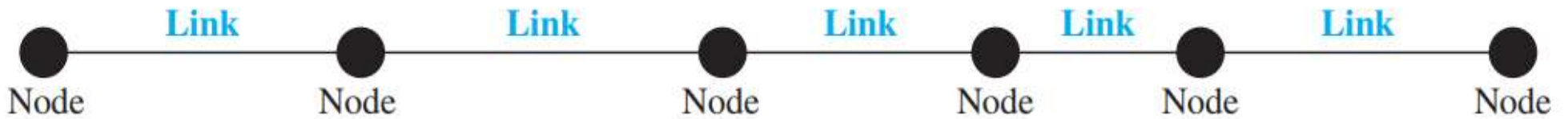
Framing, HDLC, PPP, Error detection, Error Correction, MAC Protocols, Reliable Transmission, Ethernet, 802.3, 802.5, 802.11.

Data-Link Layer

Nodes and Links



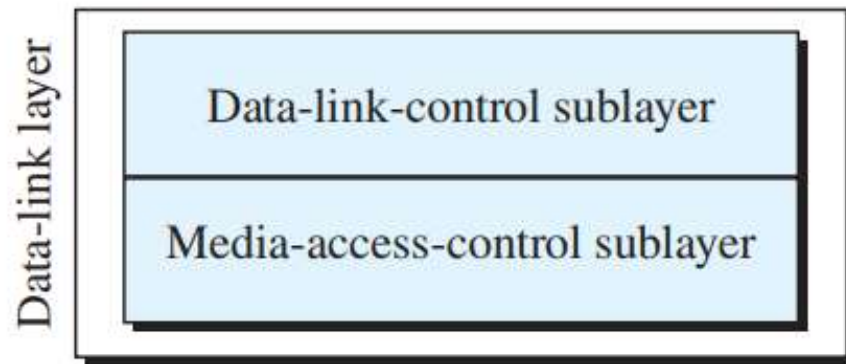
a. A small part of the Internet



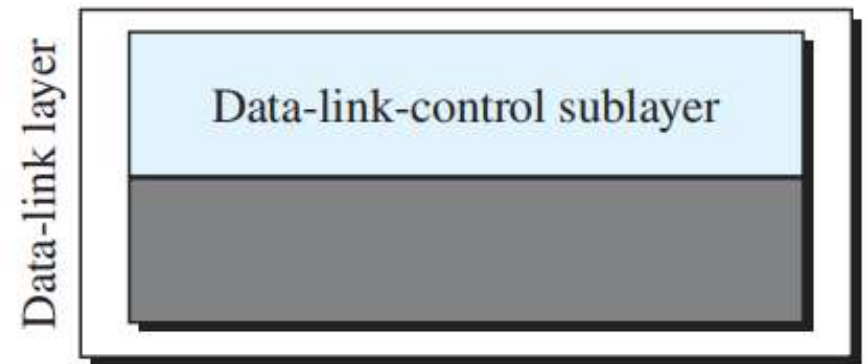
b. Nodes and links

- Point-to-Point link
- Broadcast link
- In a point-to-point link, the link is dedicated to the two devices
- In a broadcast link, the link is shared between several pairs of devices

Two Sublayers



a. Data-link layer of a broadcast link



b. Data-link layer of a point-to-point link

DATA-LINK CONTROL (DLC)

- Deals with procedures for communication between two adjacent nodes—node-to-node communication—no matter whether the link is dedicated or broadcast.
- Its functions include **framing** and **error control**.

Framing

- The data-link layer needs to pack bits into frames, so that each frame is distinguishable from another.
- Framing in the data-link layer separates a message from one source to a destination by adding a sender address and a destination address.
- The destination address defines where the packet is to go; the sender address helps the recipient acknowledge the receipt.

- Although the whole message could be packed in one frame, that is not normally done.
- One reason is that a frame can be very large, making flow and error control very inefficient.
- When a message is carried in one very large frame, even a single-bit error would require the retransmission of the whole frame.
- When a message is divided into smaller frames, a single-bit error affects only that small frame.

Generic Frame Structure



- Header: Contains control information such as addresses and sequence numbers.
- Payload: Actual data coming from the Network Layer.
- Trailer: Contains error-detection bits (e.g., checksum or CRC).

Typical Fields in a Frame



- **Flag:** A special bit pattern (e.g., 01111110) indicating the start and end of the frame.
- **Address:** Identifies the receiver on the link (used in multipoint links).
- **Control:** Manages flow and error control (e.g., sequence and acknowledgment numbers).
- **Payload (Data):** Contains the actual message or packet from the upper layers.
- **FCS (Frame Check Sequence):** Usually CRC, used to detect errors during transmission.
 - ✓ The FCS is calculated using a Cyclic Redundancy Check (CRC) algorithm on the frame content.
 - ✓ The sender appends this value in the trailer.
 - ✓ The receiver recalculates the CRC on received data and compares it with the received FCS.
 - ✓ If they don't match, the frame is assumed to be corrupted and discarded or a retransmission is requested (depending on protocol).

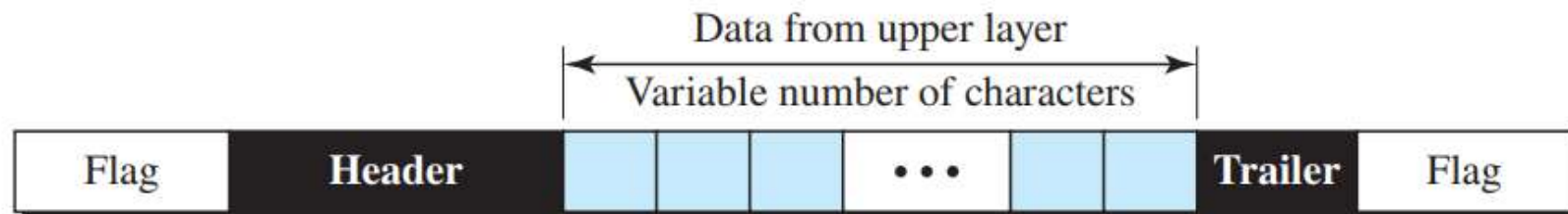
Frame Size

- Frames can be of **fixed** or **variable** size.
- In **fixed-size framing**, there is no need to define the boundaries of the frames; the size itself can be used as a delimiter.
- In **variable-size framing**, a way to define the end of one frame and the beginning of the next is needed.
- Two approaches have been used for this purpose:
 - ✓ A Character-Oriented approach
 - ✓ A Bit-Oriented approach.

Character-Oriented Framing

- In character-oriented (or byte-oriented) framing, data to be carried are 8-bit characters from a coding system such as ASCII.
- The header, which normally carries the source and destination addresses and other control information, and the trailer, which carries error detection redundant bits, are also multiples of 8 bits.
- To separate one frame from the next, an 8-bit (1-byte) flag is added at the beginning and the end of a frame.
- The flag, composed of protocol-dependent special characters, signals the start or end of a frame.

A frame in a character-oriented protocol

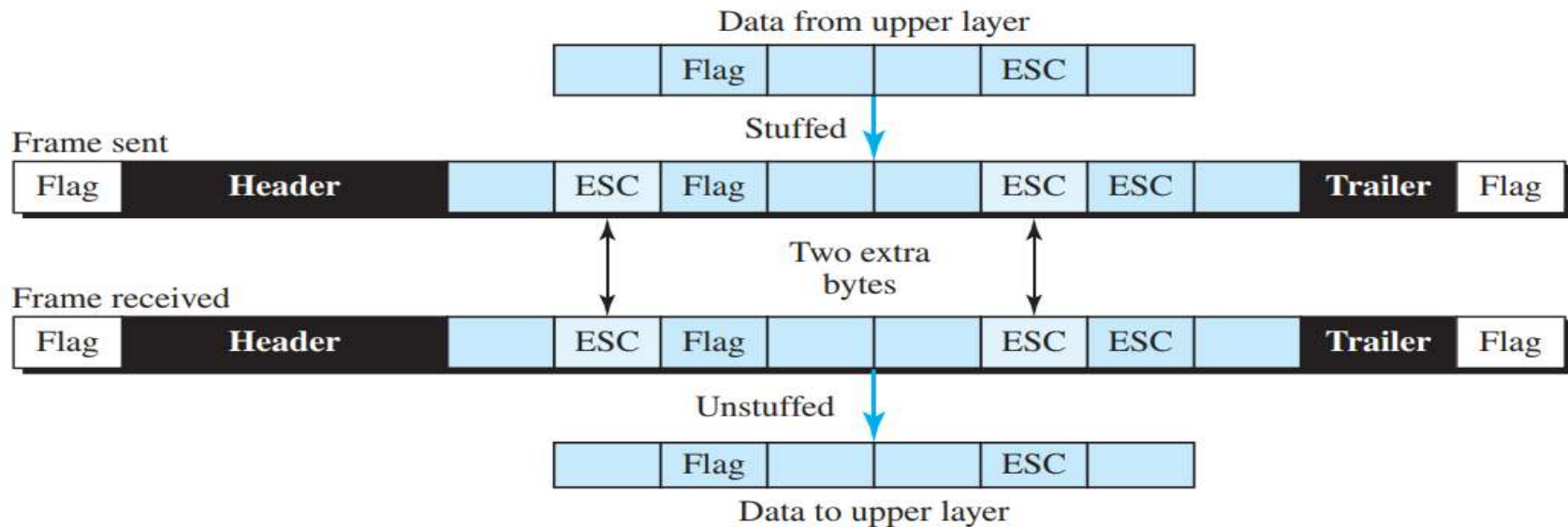


Problem

- While sending other types of information such as graphs, audio, and video; any pattern used for the flag could also be part of the information.
- If this happens, the receiver, when it encounters this pattern in the middle of the data, thinks it has reached the end of the frame.

Solution

- **Byte-Stuffing** strategy using a byte called *Escape character* (ESC), which has a predefined bit-pattern.



Other common frame delimiters: **STX (Start of Text, 0x02)** and **ETX (End of Text, 0x03)**.

Frame Structure with example

Each frame starts with STX and ends with ETX.

Frame = [STX] Data Bytes [ETX]

Example: Data to send: V I T [ETX] A P

Framed with stuffing: [STX] V I T [ESC][ETX] A P [ETX]

The receiver removes ESC during reception to recover original data.

Advantages

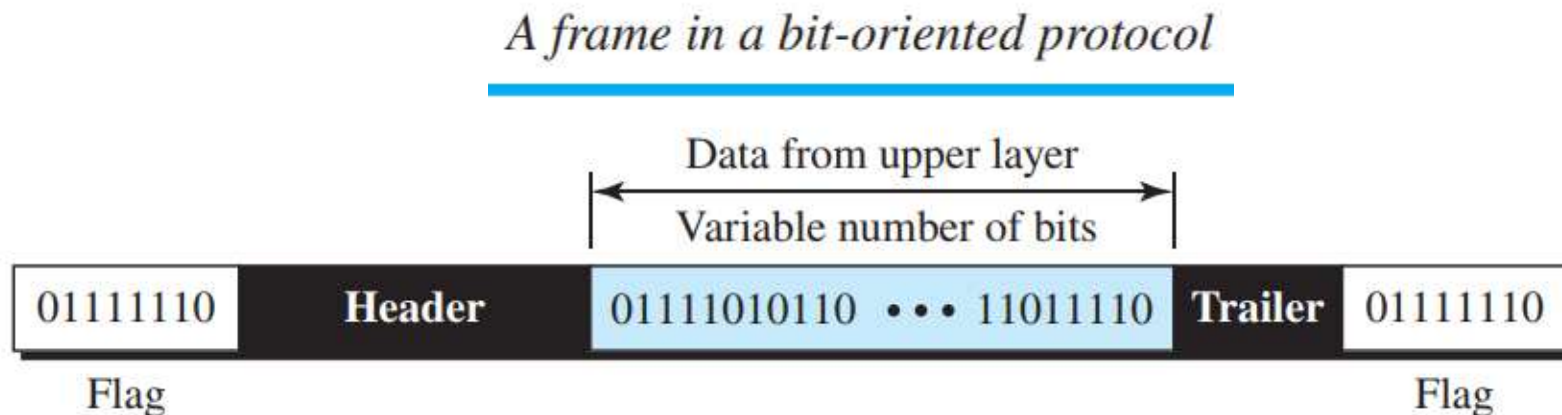
- Simple to implement.
- Suitable for text-based communication.

Disadvantages

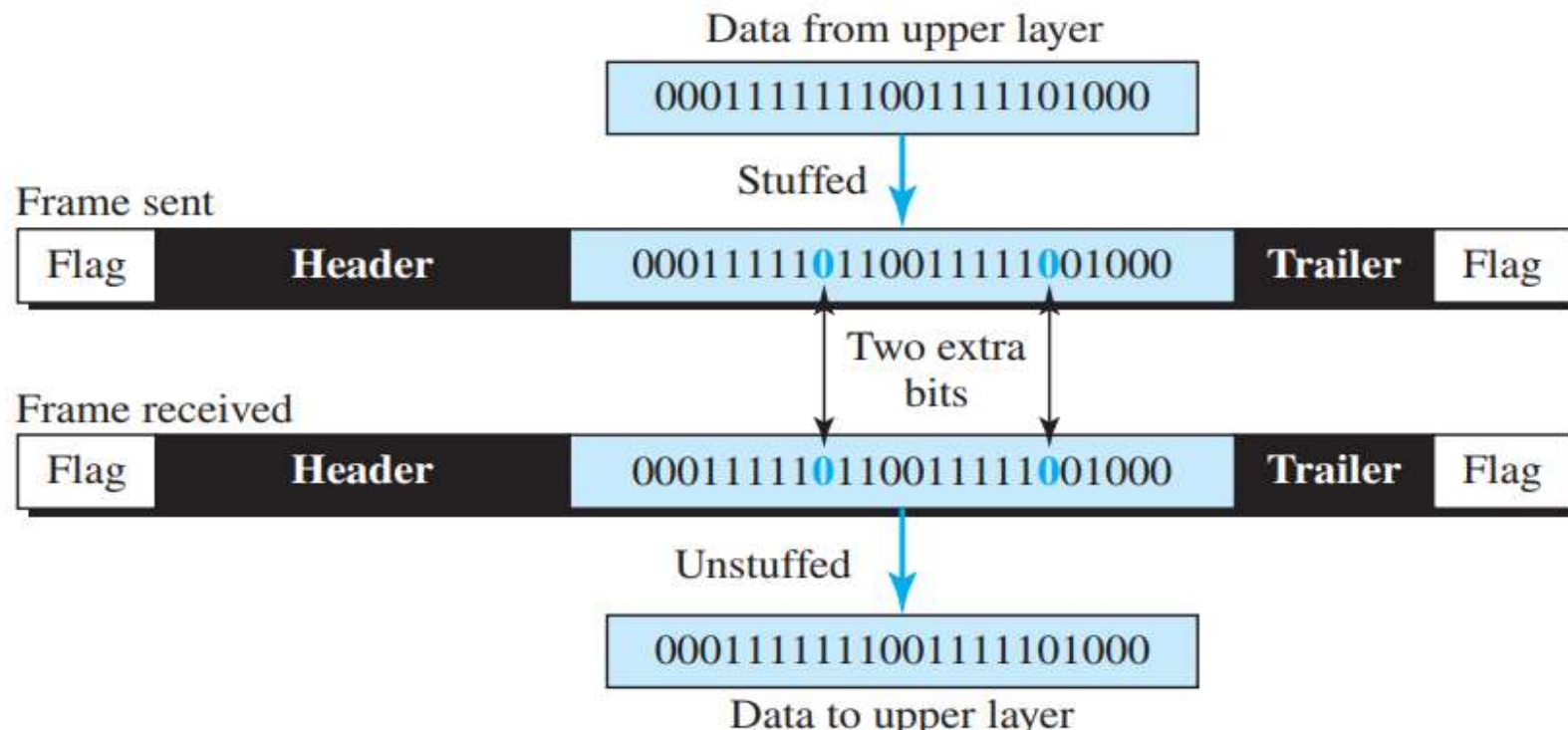
- Increased overhead due to stuffing.
- Not efficient for binary data.
- Limited usage in modern high-speed networks.

Bit-Oriented Framing

- In bit-oriented framing, in addition to headers (and possible trailers), a delimiter to separate one frame from the other is needed.
- Most protocols use a special 8-bit pattern flag, **01111110**, as the delimiter to define the beginning and end of the frame.



Bit stuffing - In bit stuffing, if a 0 and five consecutive 1 bits are encountered, an extra 0 is added.

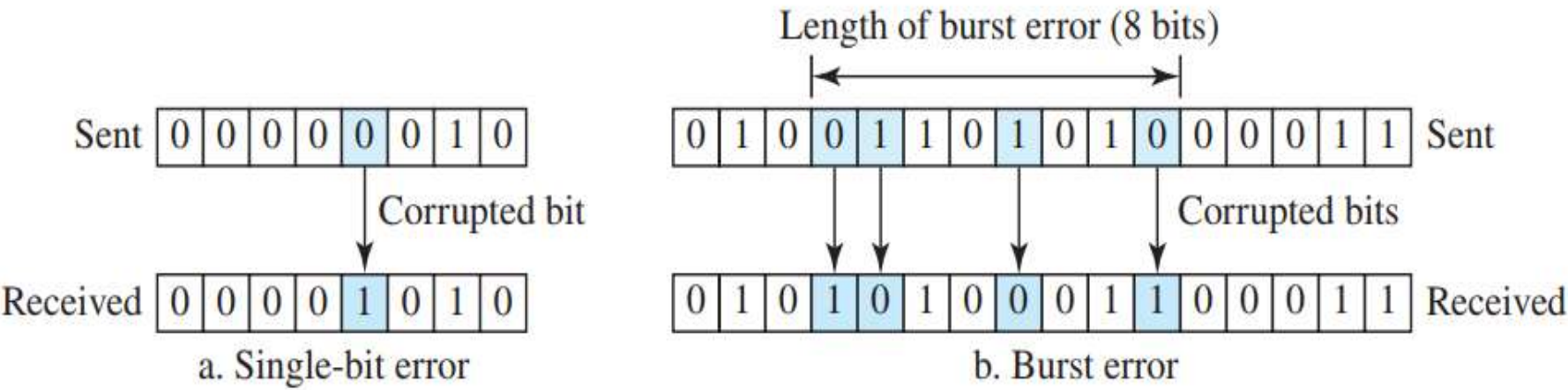


Error Control

- In the data-link layer, the term error control refers primarily to methods of error detection and retransmission (error correction is done using retransmission of the corrupted frame).
- Error control is both **error detection** and **error correction**.
- It allows the receiver to inform the sender of any frames lost or damaged in transmission and coordinates the retransmission of those frames by the sender.

Types of Errors

Single-bit error and burst error



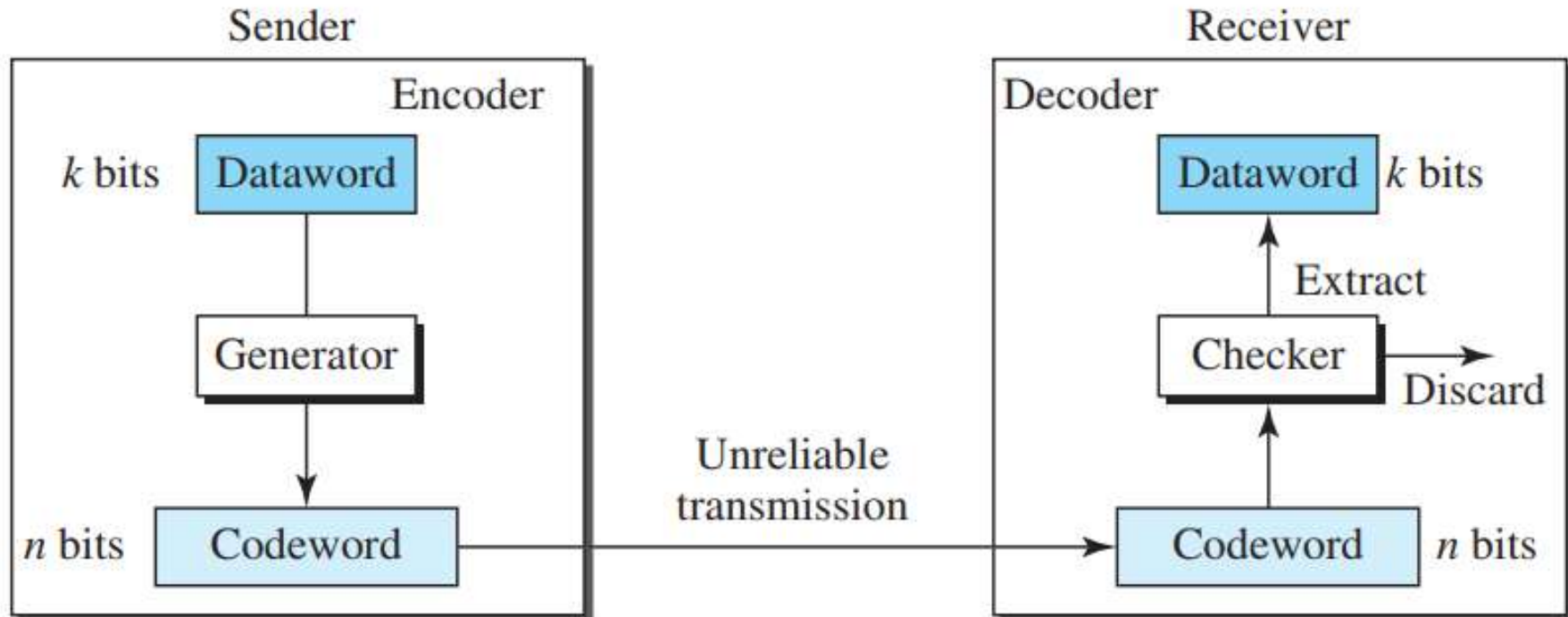
- Detection versus Correction
- Redundancy
- Coding

Block Coding

- In block coding, divide the message into blocks, each consisting of **k** bits, called **Datawords**.
- Add **r** redundant bits to each block to make the length **n = k + r**.
- The resulting **n-bit** blocks are called **Codewords**.
- The block-coding process is one-to-one.
- This means that there are **$2^n - 2^k$** codewords that are not used.
→ Are called **Invalid Codewords**.

Error Detection

1. The receiver has (or can find) a list of valid **codewords**.
2. The original **codeword** has changed to an invalid one.



Let us assume that $k = 2$ and $n = 3$. Table shows the list of datawords and codewords. Later, we will see how to derive a codeword from a dataword.

<i>Datawords</i>	<i>Codewords</i>	<i>Datawords</i>	<i>Codewords</i>
00	000	10	101
01	011	11	110

Hamming Distance

- The Hamming distance between two words (of the same size) is the number of differences between the corresponding bits.
- The Hamming distance between two words x and y is denoted as $d(x, y)$.
- Can easily be found by applying the **XOR** operation ($\oplus$) on the two words and count the number of **1s** in the result.

Minimum Hamming Distance for Error Detection

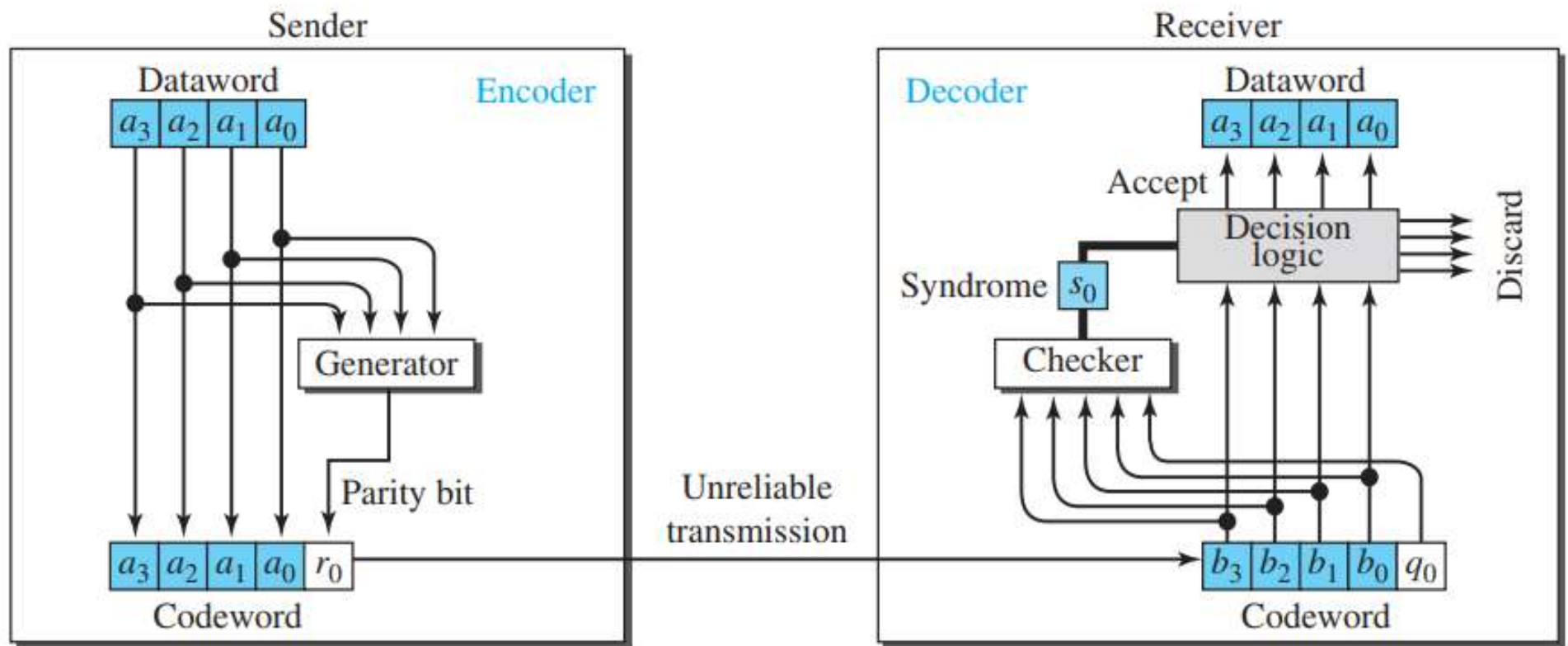
To guarantee the detection of up to s errors in all cases, the minimum Hamming distance in a block code must be $d_{\min} = s + 1$.

Parity-Check Code

Simple parity-check code $C(5, 4)$

<i>Dataword</i>	<i>Codeword</i>	<i>Dataword</i>	<i>Codeword</i>
0000	00000	1000	10001
0001	00011	1001	10010
0010	00101	1010	10100
0011	00110	1011	10111
0100	01001	1100	11000
0101	01010	1101	11011
0110	01100	1110	11101
0111	01111	1111	11110

Encoder and decoder for simple parity-check code



Cyclic Codes

- Cyclic codes are special linear block codes with one extra property.
- In a cyclic code, if a **codeword** is cyclically shifted (rotated), the result is another **codeword**.

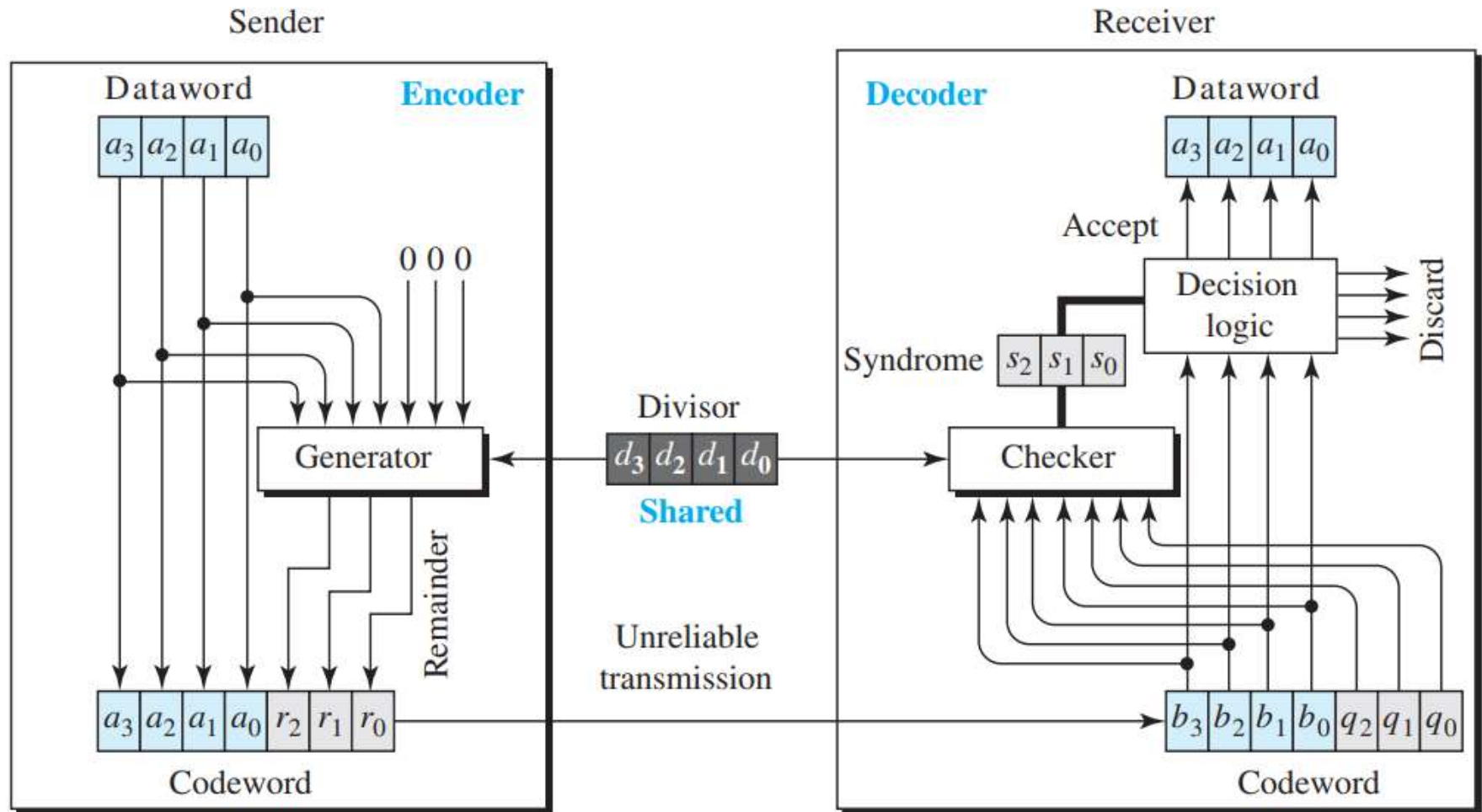
Cyclic Redundancy Check (CRC)

- Create cyclic codes to correct errors.

A CRC code with $C(7, 4)$

<i>Dataword</i>	<i>Codeword</i>	<i>Dataword</i>	<i>Codeword</i>
0000	0000000	1000	1000101
0001	0001011	1001	1001110
0010	0010110	1010	1010011
0011	0011101	1011	1011000
0100	0100111	1100	1100010
0101	0101100	1101	1101001
0110	0110001	1110	1110100
0111	0111010	1111	1111111

CRC encoder and decoder



Divisor

- The number in the name of the divisor refers to the degree of the polynomial (the highest power) representing the divisor.
- The number of bits is always one more than the degree of the polynomial.
 - For example, CRC-8 has 9 bits and CRC-32 has 33 bits.

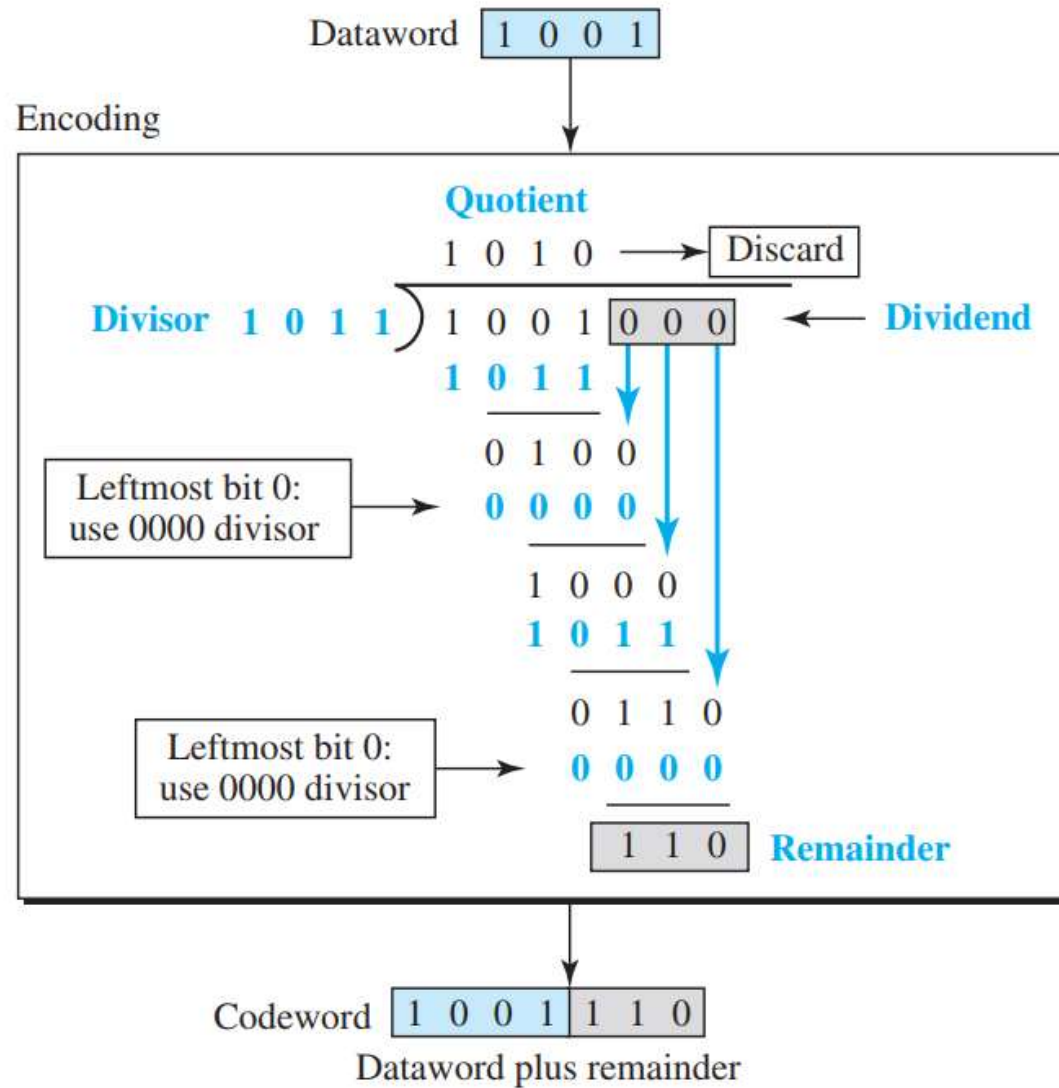
Requirement

- It is required that a bit pattern needs to have at least two properties to be considered a generator (divisor):
 1. The pattern should have at least 2 bits.
 2. The rightmost and leftmost bits should both be 1s.

Standard polynomials

<i>Name</i>	<i>Binary</i>	<i>Application</i>
CRC-8	100000111	ATM header
CRC-10	11000110101	ATM AAL
CRC-16	10001000000100001	HDLC
CRC-32	100000100110000010001110110110111	LANs

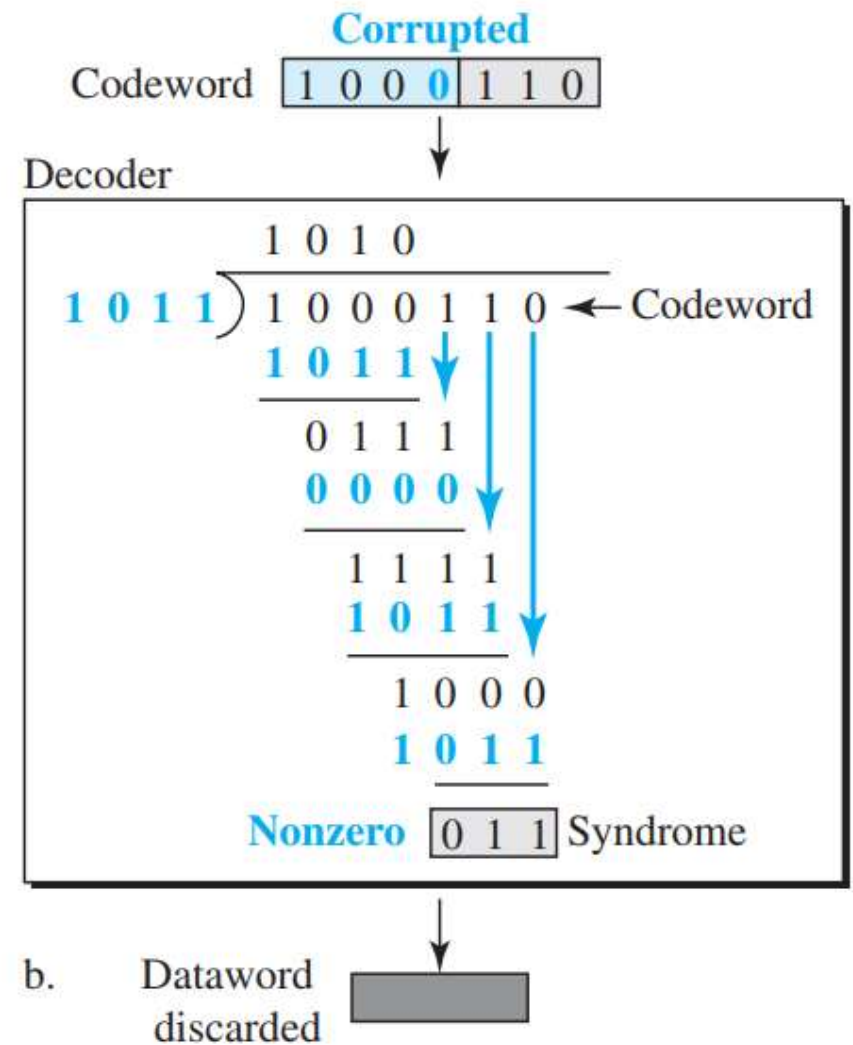
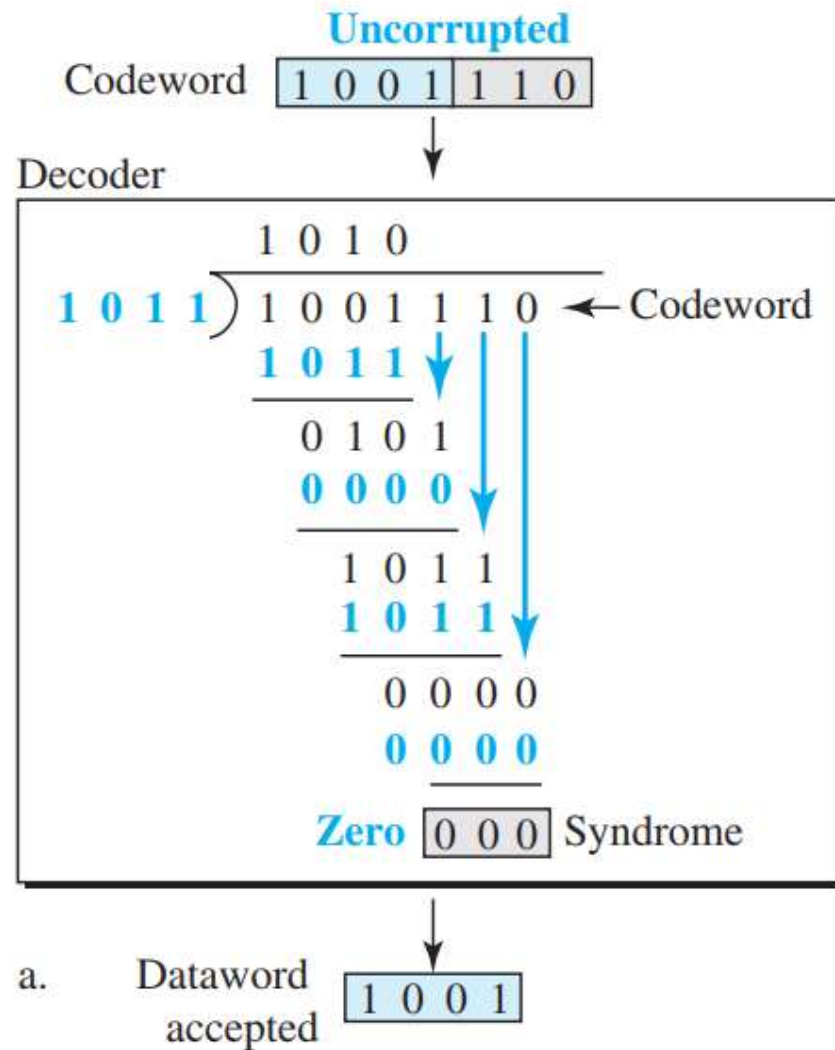
- In the **encoder**, the dataword has k bits (4 here); the codeword has n bits (7 here).
- The size of the dataword is augmented by adding $n - k$ (3 here) 0s to the right-hand side of the word.
- The n -bit result is fed into the generator. The generator uses a **divisor of size $n - k + 1$** (4 here), predefined and agreed upon.
- The generator divides the augmented dataword by the divisor (modulo-2 division).
- The quotient of the division is discarded; the remainder ($r_2 r_1 r_0$) is appended to the dataword to create the codeword.



Note:

Multiply: AND
Subtract: XOR

- The **decoder** receives the codeword (possibly corrupted in transition).
- A copy of all n bits is fed to the checker, which is a replica of the generator.
- The remainder produced by the checker is a syndrome of $n - k$ (3 here) bits, which is fed to the decision logic analyzer.
- The analyzer has a simple function.
- If the syndrome bits are all 0s, the 4 leftmost bits of the codeword are accepted as the dataword (interpreted as no error); otherwise, the 4 bits are discarded (error).



Performance

- **Single errors** - All qualified generators (see above) can detect any single-bit error.
- **Odd number of errors** - All qualified generators can detect any odd number of errors if the generator can be evenly divided by $(11)_2$ using binary division in modulo-2 arithmetic; otherwise, only some odd number errors will be detected.
- **Burst errors** - If we assume the length of the burst error is L bits and r is the length of the remainder (r is the length of the generator minus 1; it is also the value of the highest power in the polynomial representing the generator):
 - ✓ All burst errors of the size $L \leq r$ are detected.
 - ✓ All burst errors of the size $L = r + 1$ are detected with probability $1 - (0.5)^{r-1}$.
 - ✓ All burst errors of the size $L > r + 1$ are detected with probability $1 - (0.5)^r$.

Checksum:

- To detect the errors

Hamming code for Error detection and correction

Already discussed in class

DLC protocols

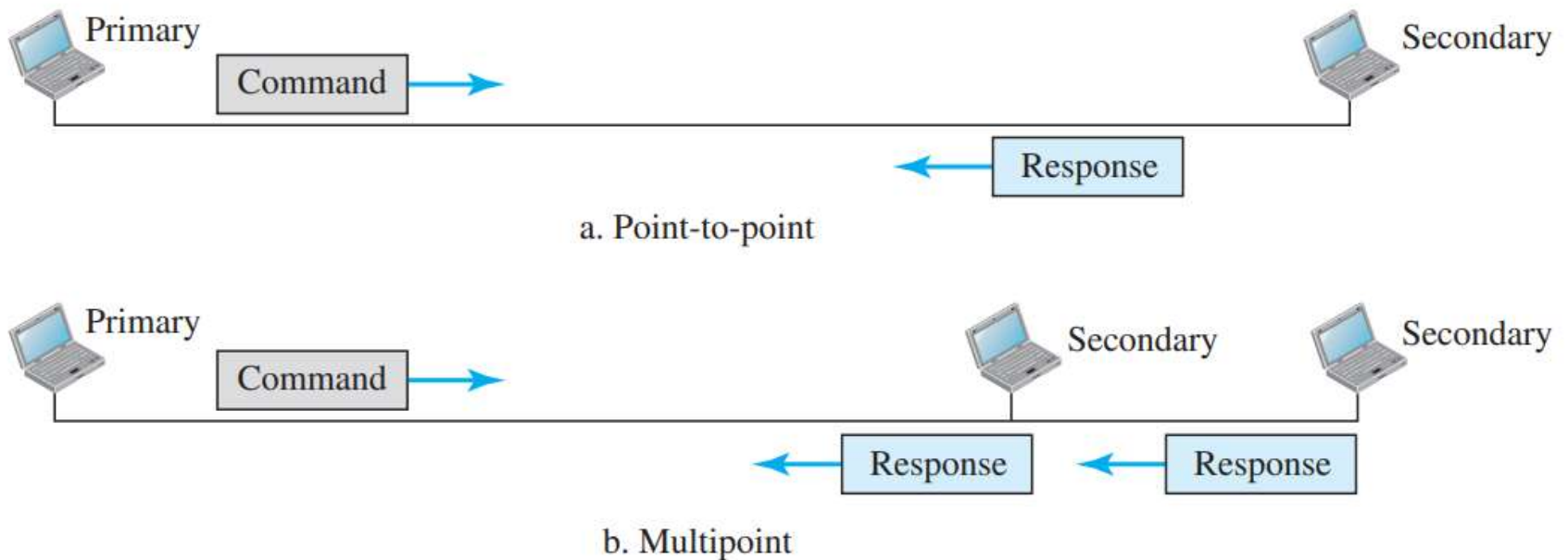
- Implements the functionalities of DLC sublayer
- Two protocols
 1. High-Level Data-Link Control (HDLC)
 2. Point-to-Point Protocol (PPP)

High-level Data-Link Control (HDLC)

- A bit-oriented protocol for communication over point-to-point and multipoint links.
- Have different configurations, but have two transfer modes.
- Transfer modes
 - Normal Response Mode (NRM)
 - Asynchronous Balanced Mode (ABM)

Normal Response Mode (NRM)

- The station configuration is **unbalanced**.
- Have one primary station and multiple secondary stations.
 - A **primary** station can *send commands*; a **secondary** station can only *respond*.
- Used for both **point-to-point** and **multipoint** links



Asynchronous Balanced Mode (**ABM**)

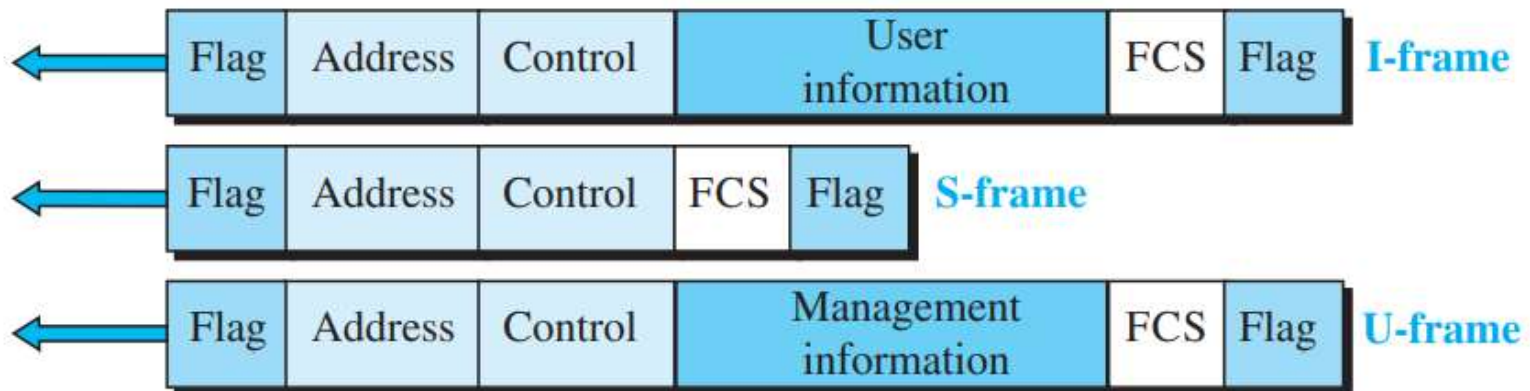
- The configuration is **balanced**
- The link is **point-to-point**
- Each station can function as a ***primary*** and a ***secondary*** (Like peers)



Frames

HDLC defines three types of frames

- Information frames (**I-frames**)
 - Used to transport user data and control information relating to user data (**piggybacking**)
- Supervisory frames (**S-frames**)
 - S-frames are used only to transport control information.
- Unnumbered frames (**U-frames**)
 - U-frames are reserved for system management.
 - Information carried by U-frames is intended for managing the link itself.



Frame --- fields and their use

- **Flag field**
 - ✓ Contains synchronization pattern 01111110, which identifies both the beginning and the end of a frame.
- **Address field**
 - ✓ Contains the address of the secondary station.
 - If a **primary station** created the frame, it contains a **to** address.
 - If a **secondary station** creates the frame, it contains a **from** address.
 - ✓ The address field can be 1 byte or several bytes long, depending on the needs of the network.
- **Control field**
 - ✓ Is of 1 or 2 bytes and used for flow and error control.
 - ✓ The control field determines the type of frame and defines its functionality.

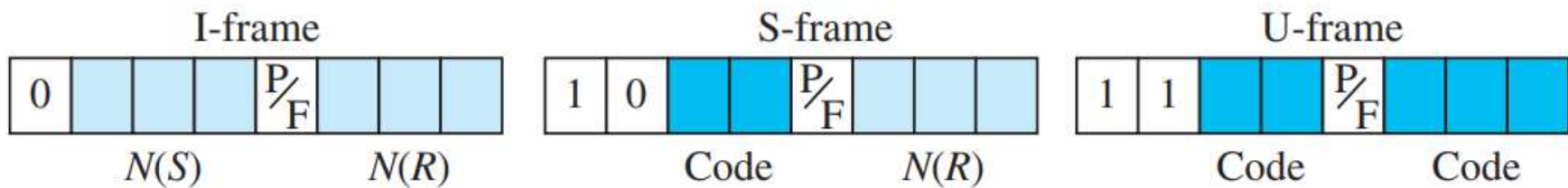
- **Information field**

- ✓ Contains the user's data from the network layer or management information.
- ✓ Its length can vary from one network to another.

- **FCS field**

- ✓ The Frame Check Sequence (FCS) is the HDLC error-detection field.
- ✓ It can contain either a 2- or 4-byte CRC.

Control field format for the different frame types



Control Field for I-Frames

I-frames are designed to carry user data from the network layer.
They can also include flow- and error-control information (piggybacking).

- The first bit defines the type.
 - ✓ If the first bit of the control field is 0, this means the frame is an **I-frame**.
- The next 3 bits, called **N(S)**, define the sequence number of the frame. With 3 bits, a sequence number between 0 and 7 can be defined.
- The last 3 bits, called **N(R)**, correspond to the acknowledgment number when piggybacking is used.
- The single bit between N(S) and N(R) is called the **P/F bit**.
 - ✓ The P/F field is a single bit with a dual purpose. It ***has meaning only when it is set (bit = 1)*** and can mean poll or final.
 - ✓ It means **poll** when the frame is sent by a primary station to a secondary station (when the address field contains the address of the receiver).
 - ✓ It means **final** when the frame is sent by a secondary station to a primary station (when the address field contains the address of the sender).

Control Field for S-Frames

Used for flow and error control whenever piggybacking is either impossible or inappropriate.
S-frames do not have information fields.

- If the first 2 bits of the control field are **10**, this means the frame is an **S-frame**.
 - ✓ The last 3 bits, called $N(R)$, correspond to the acknowledgment number (ACK) or negative acknowledgment number (NAK) depending on the type of S-frame.
- The 2 bits called **code** are used to define the type of S-frame itself.
 - ✓ With 2 bits, four types of S-frames are possible:
 - ❑ **Receive ready (RR)**
 - ❖ If the value of the code subfield is **00**, it is an **RR S-frame**.
 - ❖ This kind of frame acknowledges the receipt of a safe and sound frame or group of frames.
 - ❖ In this case, the value of the **$N(R)$ field defines the acknowledgment number**.

❑ Receive Not Ready (RNR)

- ❖ If the value of the code subfield is **10**, it is an **RNR S-frame**.
- ❖ This kind of frame is an RR frame with additional functions.
- ❖ It acknowledges the receipt of a frame or group of frames, and it announces that the *receiver is busy* and *cannot receive more frames*.
- ❖ It acts as a kind of congestion-control mechanism by asking the sender to slow down.
- ❖ The value of N(R) is the acknowledgment number.

❑ Reject (REJ)

- ❖ If the value of the code subfield is **01**, it is an **REJ S-frame**.
- ❖ The frame is rejected.

❑ Selective reject (SREJ)

- ❖ If the value of the code subfield is **11**, it is an **SREJ S-frame**.
- ❖ The receiver can keep or delete the frame.

Control Field for U-Frames

- Unnumbered frames are used to exchange session management and control information between connected devices.
- Unlike S-frames, U-frames contain an information field, but ***used for system management information, not user data.***
- As with S-frames, however, much of the information carried by U-frames is contained in codes included in the control field.
- U-frame codes are divided into two sections:
 - ✓ A 2-bit prefix before the P/F bit
 - ✓ A 3-bit suffix after the P/F bit
- Together, these two segments (5 bits) can be used to create up to 32 different types of U-frames.

Point-to-Point Protocol (PPP)

- One of the most common protocols for point-to-point access
- Majority of these users have a traditional modem; they are connected to the Internet through a telephone line, which provides the services of the physical layer.
- To control and manage the transfer of data, there is a need for point-to-point access at the data-link layer.

Services Provided by PPP

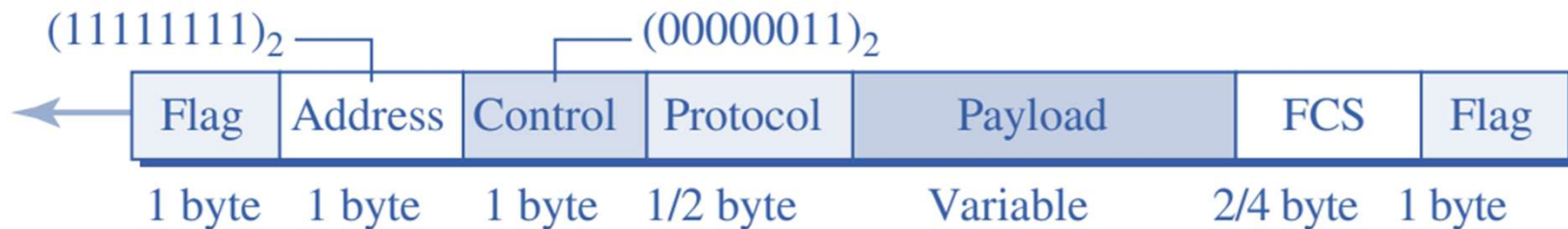
- PPP defines the format of the frame to be exchanged between devices.
- It also defines how two devices can negotiate the establishment of the link and the exchange of data.
- PPP is designed to accept payloads from several network layers.
- **Authentication** is also provided in the protocol, but it is optional.
- The new version of PPP, called **Multilink PPP**, provides connections over multiple links.
- One interesting feature of PPP is that it provides **network address configuration**.
- This is particularly useful when a home user needs a temporary network address to connect to the Internet.

Services Not Provided by PPP

- PPP does not provide flow control. A sender can send several frames one after another with no concern about overwhelming the receiver.
- PPP has a very simple mechanism for error control. A CRC field is used to detect errors. If the frame is corrupted, it is silently discarded; the upper-layer protocol needs to take care of the problem.
- Lack of error control and sequence numbering may cause a packet to be received out of order.
- PPP does not provide a sophisticated addressing mechanism to handle frames in a multipoint configuration.

Framing

- PPP uses a character-oriented frame.



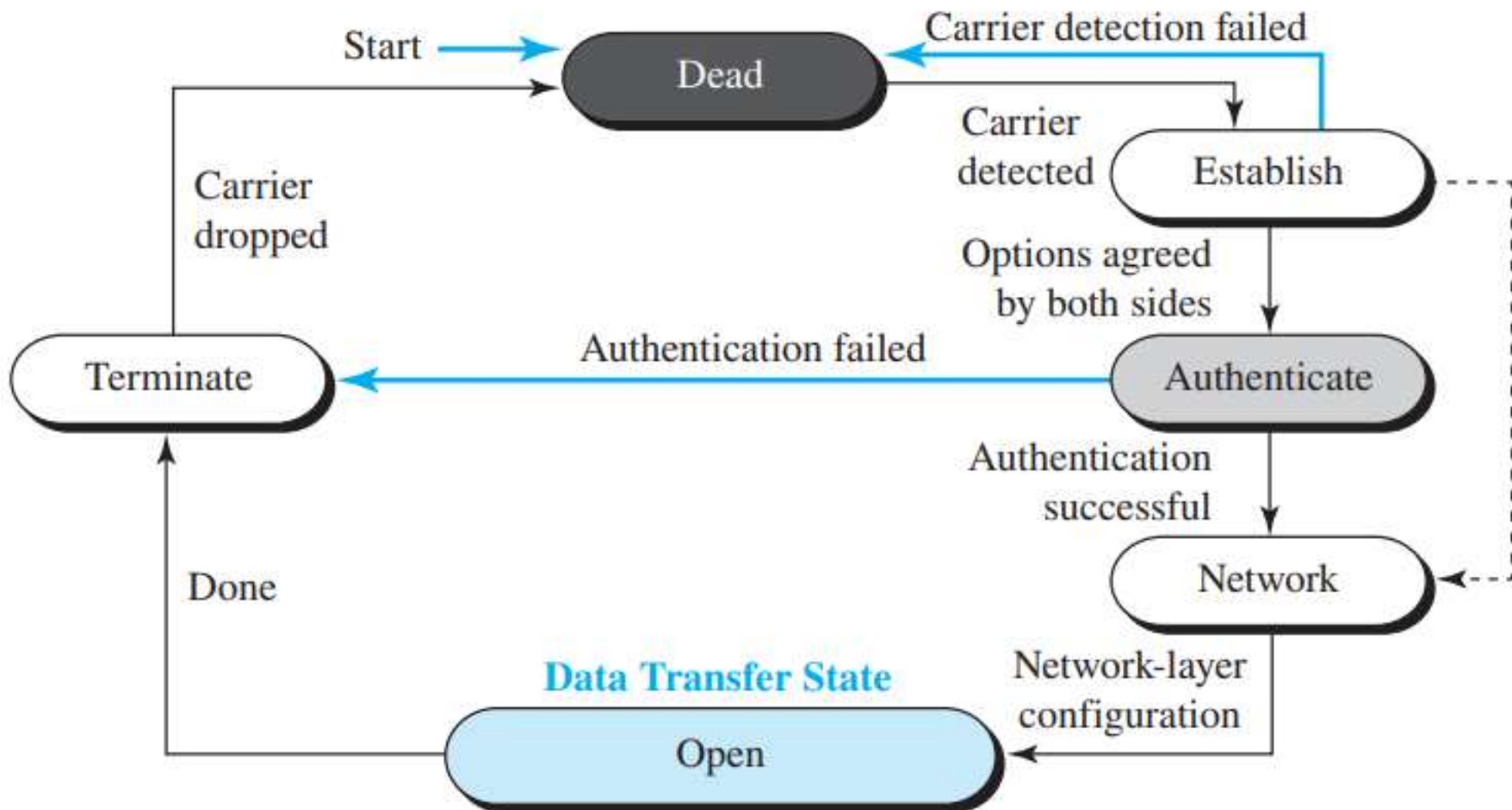
- ✓ **Flag** - A PPP frame starts and ends with a 1-byte flag with the bit pattern 01111110.
- ✓ **Address** - The address field in this protocol is a constant value and set to 11111111 (broadcast address).
- ✓ **Control** - This field is set to the constant value 00000011 (similar to U-frames in HDLC). PPP does not provide any flow control. Error control is also limited to error detection.

- ✓ **Protocol** - The protocol field defines what is being carried in the data field: either user data or other information. This field is by default 2 bytes long, but the two parties can agree to use only 1 byte.
- ✓ **Payload field** - This field carries either the user data or other information. The data field is a sequence of bytes with the default of a maximum of 1500 bytes, but this can be changed during negotiation. The data field is byte-stuffed if the flag byte pattern appears in this field. Because there is no field defining the size of the data field, padding is needed if the size is less than the maximum default value or the maximum negotiated value.
- ✓ **FCS** - The frame check sequence (FCS) is simply a 2-byte or 4-byte standard CRC.

Byte Stuffing

- As PPP is a byte-oriented protocol, the flag in PPP is a byte that needs to be escaped whenever it appears in the data section of the frame.
- The escape byte is 01111101.

Transition Phases



Multiplexing

Three sets of protocols are defined to make PPP powerful:

- One Link Control Protocol (LCP)
- Two Authentication Protocols (APs)
- Several Network Control Protocols (NCPs)

Legend

LCP : Link control protocol
AP : Authentication protocol
NCP: Network control protocol

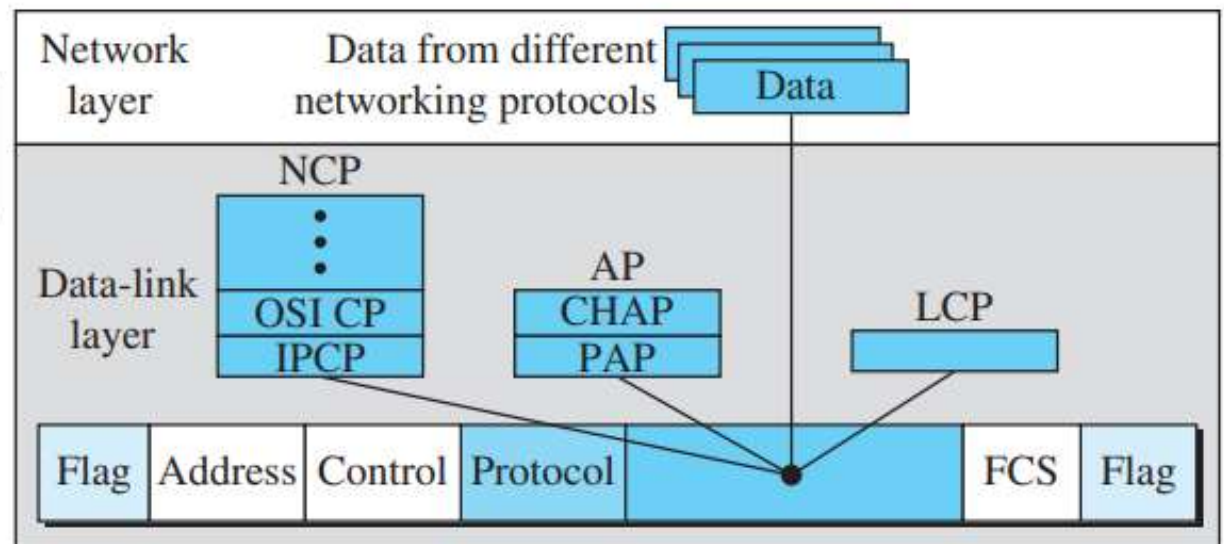
Protocol values:

LCP: 0xC021

AP : 0xC023 and 0xC223

NCP: 0x8021 and

Data: 0x0021 and



Link Control Protocol

- The Link Control Protocol (LCP) is responsible for establishing, maintaining, configuring, and terminating links.
- It also provides negotiation mechanisms to set options between the two endpoints.
- Both endpoints of the link must reach an agreement about the options before the link can be established.

Authentication Protocols

- Authentication means validating the identity of a user who needs to access a set of resources.
- PPP has created two protocols for authentication:
 - ✓ Password Authentication Protocol (PAP)
 - ✓ Challenge Handshake Authentication Protocol (CHAP)

Network Control Protocols

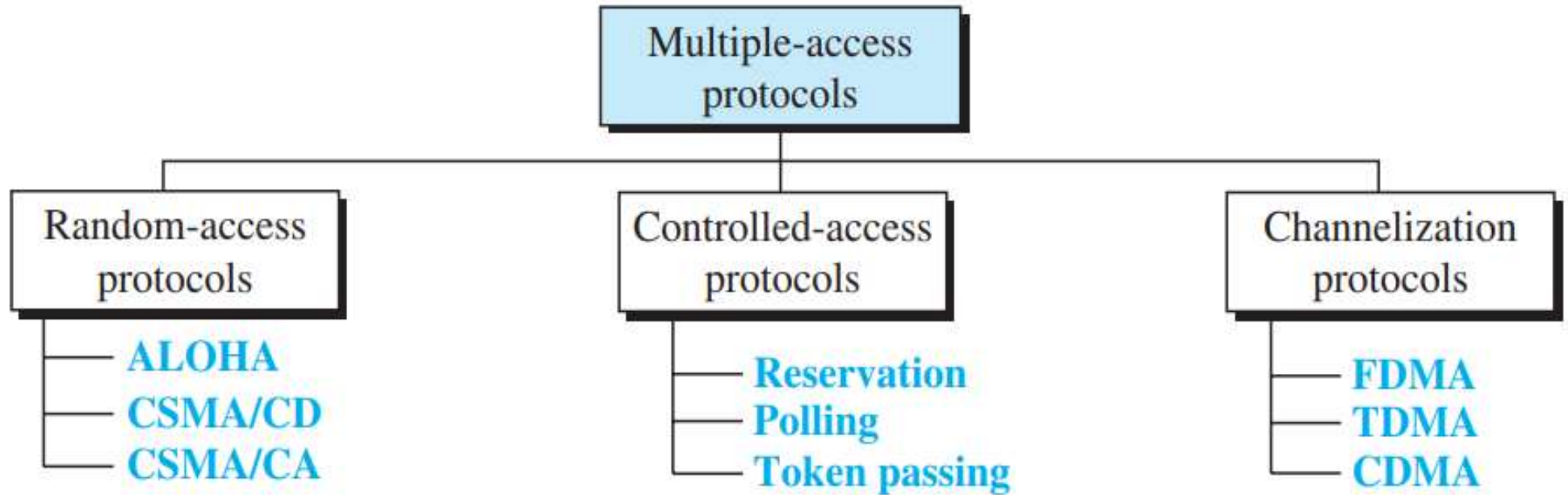
- PPP is a multiple-network-layer protocol.
- One NCP protocol is the Internet Protocol Control Protocol (IPCP). This protocol configures the link used to carry IP data packets in the Internet.

Data from the Network Layer

- After the network-layer configuration is completed by one of the NCP protocols, users can exchange data packets from the network layer.
- Here again, there are different protocol fields for different network layers. For example,
 - ✓ If PPP is carrying data from the IP network layer, the field value is $(0021)_{16}$.
 - ✓ If PPP is carrying data from the OSI network layer, the protocol field value is $(0023)_{16}$.

Media Access Control (MAC)

- One of the two sub-layers in Data-Link Layer.
- When nodes or stations are connected and use a common link, called a **multipoint** or **broadcast** link, a multiple-access protocol to coordinate access to the link is needed.
- Need to be sure that each node gets access to the link.
- To prevent any collision between nodes. To handle the collision if it occurs.
- Many protocols have been devised to handle access to a shared link.



Random Access

- Also called “Contention method”.
- In this, no station is superior to another station and none is assigned control over another.
- At each instance, a station that has data to send uses a procedure defined by the protocol to make a decision on whether or not to send.
- This decision depends on the state of the medium (idle or busy).

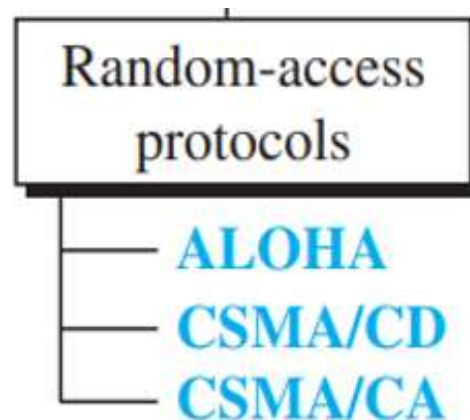
Why named 'Random'?

- There is no scheduled time for a station to transmit.
- Transmission is random among the stations.

Why named 'Contention'?

- No rules specify which station should send next.
- Stations compete with one another to access the medium.

- In this random-access method, each station has the right to the medium without being controlled by any other station.
- But, if more than one station tries to send, there is an access conflict—**collision**—and the *frames will be either destroyed or modified*.
- Many protocols were suggested to handle this.



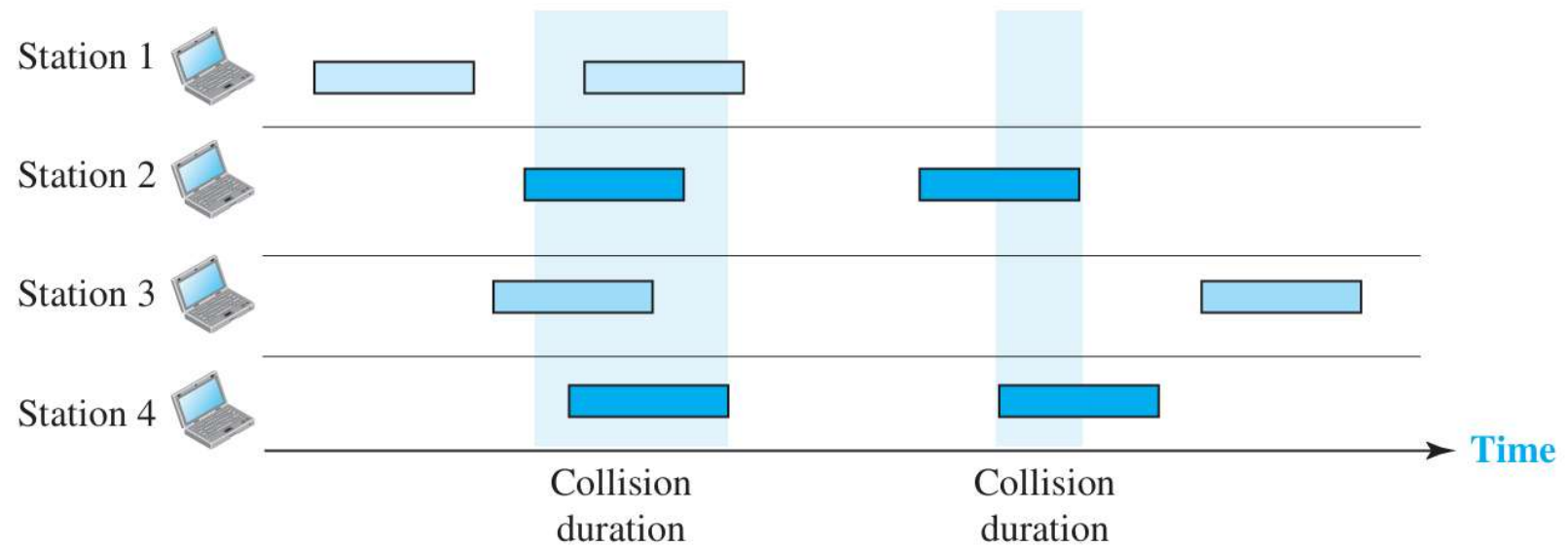
- One such protocol is the '**ALOHA**' - used a very simple procedure called ***Multiple Access (MA)***.
- ***Carrier Sense Multiple Access (CSMA)*** - a procedure that forces the station to sense the medium before transmitting.
- ***Carrier Sense Multiple Access with Collision Detection (CSMA/CD)*** - tells the station what to do when a collision is detected.
- ***Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA)*** - tries to avoid the collision.

ALOHA, the first random-access method, was developed at the University of Hawaii in early 1970.

Pure ALOHA

- The original ALOHA protocol is called pure ALOHA. This is a simple but elegant protocol.
- The idea is that each station sends a frame whenever it has a frame to send (multiple access).
- As there is only one channel to share, there is the possibility of collision between frames from different stations.

Frames in a pure ALOHA network

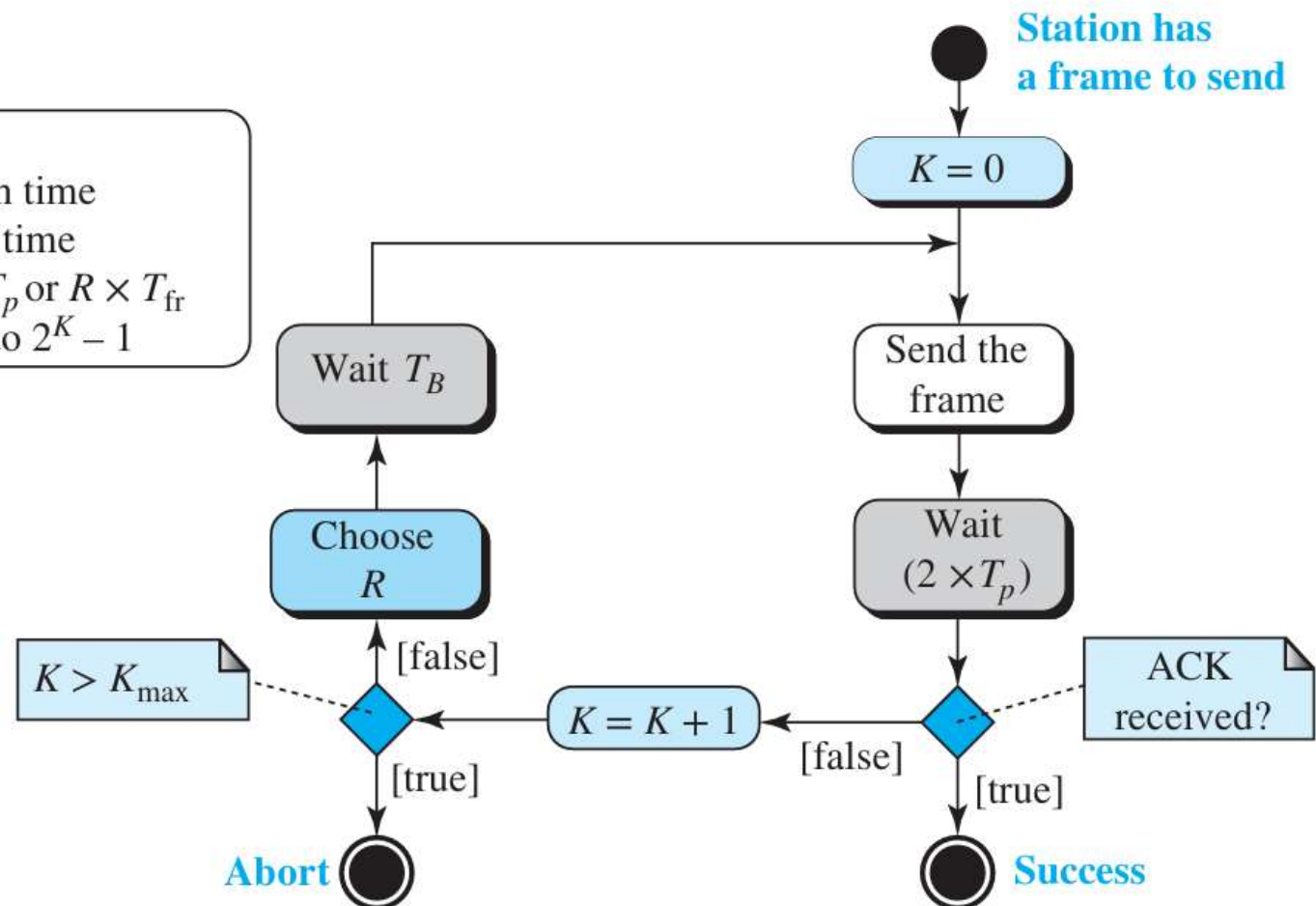


- The pure ALOHA protocol relies on acknowledgments from the receiver.
- When a station sends a frame, it expects the receiver to send an acknowledgment.
- If the acknowledgment does not arrive after a time-out period, the station assumes that the frame has been destroyed and resends the frame.
- A **collision** involves two or more stations. If all these stations try to resend their frames after the time-out, the frames will collide again.
- Pure ALOHA dictates that when the time-out period passes, each station waits a random amount of time, called **Backoff Time T_B** , before resending its frame.
- The randomness will help avoid more collisions.
- And, to prevent congesting the channel with retransmitted frames, after a maximum number of retransmission attempts **K_{max}** , a station must give up and try later.

Procedure for pure ALOHA protocol

Legend

K : Number of attempts
 T_p : Maximum propagation time
 T_{fr} : Average transmission time
 T_B : (Backoff time): $R \times T_p$ or $R \times T_{fr}$
 R : (Random number): 0 to $2^K - 1$



- The time-out period is equal to the maximum possible round-trip propagation delay, which is twice the amount of time required to send a frame between the two most widely separated stations ($2 \times T_p$).
- The **Backoff Time** T_B is a random value that normally depends on K (*the number of attempted unsuccessful transmissions*).
- The formula for T_B depends on the implementation. One common formula is the *Binary Exponential Backoff*. In this method, for each retransmission, a multiplier $R = 0$ to $2^k - 1$ is randomly chosen and multiplied by T_p (*maximum propagation time*) or T_{fr} (*the average time required to send out a frame*) to find T_B .
- Here, the range of the random numbers increases after each collision. The value of K_{max} is usually chosen as **15**.

Vulnerable time for pure ALOHA protocol

